MIL-L-51050B(ME)
13 December 1971
SUPERSEDING
MIL-L-51050A(GL)
13 December 1965

MILITARY SPECIFICATION

LABORATORY, PETROLEUM, SEMITRAILER MOUNTED

1. SCOPE

1.1 This specification covers a mobile petroleum laboratory, fully equipped with testing equipment, chemicals, and supplies, to test petroleum and associated products.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

O-E-910	- Extinguishers, Fire, Carbon-Dioxide (Hand and Wheeled Types).
TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
PPP-T-60	- Tape: Packaging, Waterproof.

Military

MIL-P-116	- Preservation, Methods of.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-B-7883	- Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.
MIL-P-26514	- Polyurethane Foam, Rigid or Flexible, for Packaging.
MIL-S-46756	- Semitrailer's Van: 10 Ton, 4-Wheel, Series.

FSC 6640

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STANDARDS

Military

MIL-STD-101	- Color Code for Pipelines and for Compressed Gas Cylinders.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-281	- Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-646	- Electrical Circuit (Wire Marking) Identification for Tactical Military Vehicles.
MS39020	- Band, Marker, Blank.

DRAWING

ME

DL13219E1600	- Laboratory, Petroleum, Semitrailer Mounted.

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASSOCIATION OF AMERICAN RAILROADS

AAR Manual, Section 1	- Loading of Commodities on Open Top Cars.
AAR Rules, Section 6	- Loading of Department of Defense Material on Open Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 59 East Van Buren Street, Chicago, Illinois 60605.)

DEPARTMENT OF TRANSPORTATION

49 CFR 171-190	- Department of Transportation Regulations for the Transportation of Hazardous Materials.

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Motor Carrier Safety Regulations

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

NATIONAL FIRE PROTECTION ASSOCIATION

National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, Massachusetts 02110.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Service.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, Two Pennsylvania Plaza, New York, N.Y. 10001.)

AMERICAN WELDING SOCIETY

Standard Qualifications Procedure

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, N.Y. 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Welding Qualifications Procedure

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, N.Y. 10017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Description. The semitrailer mounted petroleum laboratory (hereinafter called "mobile laboratory") shall be in accordance with Data List DL13219E1600 and as specified herein.

3.1.1 Drawing. The drawing forming a part of this specification is an engineering design drawing. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (first-produced mobile laboratory). The supplier shall furnish one mobile laboratory for examination and testing within the time frame specified, to prove that his production methods will produce mobile laboratories that comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and as shown on the applicable drawing. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.4 Semitrailer. When specified, the semitrailer shall be in accordance with the applicable drawing and shall conform to MIL-S-46756.

3.5 Components. Items of equipment shall be as specified in SC 6640-97-CL-E02. When specified, the items of equipment shall be furnished by the supplier and shall be located in the mobile laboratory as specified herein and on the applicable drawings (see 6.9 and 6.10).

3.6 Performance. The mobile laboratory shall be capable of being towed by an M-52, 5-ton, 6 by 6 tractor and sustain over-the-road, across curbing and rough terrain travel. The fully equipped laboratory shall be air transportable in the C-130, C-133 and C-141 USAF aircraft. Provisions shall be provided within the mobile laboratory to secure all equipment, supplies, chemicals, books, binders, and manuals. When towed or air transported as specified herein, no damage, dislocation, deformation, breakage, cracking, or leaking shall be evident.

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3.7 Cushion material. Cushion material for cabinets, drawers, and shelves shall conform to MIL-C-26514, type I, class 2, except the foam shall have a polyetherurethane molecular structure. The foam shall have 2.0 pound per square foot density and 60 pores per linear inch. Color shall be charcoal.

3.7.1 Installation. The cushion material shall be installed in all cabinets, drawers, and on all cabinet shelves, in the form as shown on the applicable drawings, to provide protection for stored equipment from impact and vibration damage.

3.8 Utility systems. The mobile laboratory shall contain the following utility systems as specified on the applicable drawings and shall not leak when tested as specified in 4.5.2.5.

(a) Water
(b) Water drainage
(c) Air
(d) Vacuum
(e) Butane/propane
(f) Oxygen

3.9 Color code marking. All utility piping and valves shall have color code marking in accordance with MIL-STD-101.

3.10 Electrically operated utility and testing equipment and systems. Electrically operated utility and testing equipment and systems shall be as specified on the applicable drawings and shall operate as intended without malfunction or without evidence of binding or overheating.

3.11 Electrical wiring. All electrical wiring shall comply with the National Electrical Code (NEC) of the National Fire Protection Association.

3.11.1 Cords. All electrical cords shall be as short as practicable while allowing sufficient slack for repair. The cords shall be installed in such a manner as to insure safe conditions for conducting tests with highly volatile materials.

3.11.2 Permanent wiring. Major components of electrical equipment shall be tied into the electrical system permanently and shall be equipped with "ON" and "OFF" switches as specified on applicable drawings.

3.11.3 Protection. Grommets shall be used to prevent chafing, rubbing, and abrading where wires or harnesses pass through any structural member. Cushioned straps shall be used to support wiring and harnesses. Maximum unsupported length shall be 36 inches.

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3.11.4 Tracing. Wiring shall be number-traced in accordance with MIL-STD-646 using markers conforming to MS39020.

3.12 Threaded parts. All screw threads shall be in accordance with National Bureau of Standards Handbook H28 and, unless otherwise specified, shall be class 2 fit.

3.13 Hazardous materials. The transportation of chemicals, compressed gasses and other hazardous materials shall comply with the Department of Transportation regulation 49 CFR 171-190 (see 6.4).

3.14 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173 with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish.

(a) Components or circuit elements that are inherently fungus and moisture resistant, or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.15 Fire extinguishers. Unless otherwise specified, the Government will furnish four fire extinguishers, carbon dioxide, FSN 4210-270-4512, conforming to O-E-910, type I, size 5. The mountings shall be as specified in the applicable drawings (see 6.5).

3.16 Identification marking. The mobile laboratory shall be identified in accordance with MIL-STD-130. The marking shall be applied to the mobile laboratory on plates conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.17 Shipping data plate. The shipping data plate shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material and, in addition, shall show the silhouette of the mobile laboratory in transport

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position indicating the center of gravity and the location and capacity of the lifting and the tiedown attachments. The plate shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.18 Instruction plates. Each mobile laboratory shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the mobile laboratory. Instruction plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.19 Treatment and painting. The portions of the mobile laboratory painted as specified on the applicable drawings shall be cleaned, treated, and painted in accordance with MIL-T-704, type A. Nonpermissible galvanic couples (dissimilar metals) as defined in MIL-T-704 shall be insulated as specified in MIL-T-704.

3.20 Instructional material. Unless otherwise specified, reference books and commercial catalogs as specified in SC 6640-97-CL-E02 shall be furnished by the supplier and shall be included with each mobile laboratory, together with technical literature, data sheets and certifications (as applicable) received by the supplier from manufacturers of the component items of equipment and supply.

3.21 Repair parts and maintenance tools. Such repair parts and maintenance tools as are specified shall be furnished.

3.22 Items to be furnished with the mobile laboratory. No shipment of mobile laboratories shall be made unless repair parts, tools, instructional material, and accessories are included with the shipment or unless approval for shipment of the mobile laboratory without such items has been received from the contracting officer.

3.23 Government-furnished property. When specified, the following property in the quantities indicated will be furnished by the Government (see 6.6):

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Item No.	Description	Identification	Qty
1.	Semitrailer, Van; Petroleum Laboratory	2330-122-4966	1
2.	Air conditioner	4120-963-4477	1
3.	Blending kit, portable	6640-738-2125	1
4.	Binders (3) containing current specifications, inspection information, special regulations and pertinent publications		1 set
5.	Binders, empty	7510-188-6951	3
6.	Books, record type	7530-222-3525	2
7.	FED. TEST METHOD STANDARDS for Testing Petroleum	FED. STD. No. 791	1
8.	Fuel Test Engine, Motor/Research Method	6640-562-5940	1
9.	Manual, Air Conditioner Parts List	TM-5-4120-295-25P	1
10.	Manual, Air Conditioner Operation and Maintenance	TM-5-4120-295-15	1
11.	Manual, Inspecting and Testing Petroleum Products	TM-10-1105	1
12.	Manual, Petroleum Handling Equipment and Operations	TM-10-1101	1
13.	Manual, Operating and Maintenance, for the Laboratory, Petroleum, Semitrailer Mounted	Manufacturer's Commercial Manual	1
14.	Supply Catalog, Laboratory Petroleum, Semitrailer Mounted	SC 6640-97-CL-E02	1
15.	Tool Kit General Mechanics	5180-629-9783	1

3.24 Vehicle weight classification. The supplier shall complete the Government-furnished vehicle weight classification forms and apply the vehicle weight classification number(s) to the mobile laboratory as specified (see 6.8).

3.25 Workmanship.

3.25.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

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3.25.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the mobile laboratory to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch, unless appropriate bridging techniques are used. Unless otherwise specified, the welding process used in fabrication of the mobile laboratory shall be at the option of the supplier.

3.25.3 Welders. Before assigning any welder to manual welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society.
Welding Qualifications of the ASME.

Suppliers who only make horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, supplier's standard welder qualification may be substituted in lieu of the above codes provided that the supplier's procedure is equivalent to the above codes. The supplier shall be responsible for determining that automatic welder equipment operators are capable of producing quality welds.

3.25.4 Welding practices.

(a) Preheat of materials being welded and maximum interpass temperatures during welding shall be in accordance with the supplier's recommendations.
(b) Tack welds shall be subject to the same quality requirements as final welds.
(c) Work shall be positioned for flat welding whenever practicable.
(d) Procedures and sequences shall be such that distortion and shrinkage will be held to a minimum. When straightening is required, caution shall be exercised to insure that the straightening process does not weaken the part and the mobile laboratory.

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3.25.5 Weld quality.

(a) The supplier shall be responsible for determining the inspections needed to insure that when the welded parts are assembled together to make the mobile laboratory, the mobile laboratory shall conform to the inspection requirements specified in section 4.
(b) All weldments shall be free of slag, flux, weld spatter, and other impurities detrimental to both the appearance and strength of the weldment.
(c) Undercut in weldments shall be held to a minimum but shall not be more than 1/32 inch deep.

3.25.6 Brazing. Brazing of steels, copper alloys and nickel alloys shall conform to MIL-B-7883.

3.25.7 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.25.8 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member and shall be in accordance with SAE J492.

3.25.9 Machine work. Tolerances and gages for metal fits shall conform to the limits specified herein and on applicable drawings and to the standards of the industry.

3.25.10 Floor. All floor joints shall be tight, without recesses or crevices where dirt and refuse can accumulate. The upper, or exposed surface, shall present a smooth even face.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) First-produced laboratory inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 First-produced laboratory.

4.3.1 Examination. The first-produced mobile laboratory shall be examined for the defects marked "X" in columns 1 and 2 of table I. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first-produced mobile laboratory shall be tested as specified in 4.5.2.1 through 4.5.2.6. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each mobile laboratory shall be examined for the defects marked "X" in column 3 of table I. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each mobile laboratory shall be tested as specified in 4.5.2.1 through 4.5.2.3 and 4.5.2.5 through 4.5.2.6. Failure of any test shall be cause for rejection.

4.5 Inspection schedule.

4.5.1 Examination. Examination shall be in accordance with table I.

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Table I. Examination schedule

First-produced				
Pre-test	Post-test	Quality conformance		Defect
1	2	3		4
X	X	X	101.	Assembly incomplete or incorrect.
X	-	-	102.	Design detail not as specified.
X	-	-	103.	Material not as specified.
X	X	X	104.	Parts and components missing or not as specified.
X	-	-	105.	Dimensions not as specified.
X	X	X	106.	Cabinets not as specified.
X	X	X	107.	Drawers not as specified.
X	-	X	108.	Cushion material not as specified.
X	X	X	109.	Cushion material not installed as specified.
X	X	X	110.	Locking devices not as specified.
X	X	X	111.	Cabinet drawers and doors identification not as specified.
X	X	X	112.	Utility systems not as specified.
X	X	X	113.	Piping not as specified.
X	X	X	114.	Piping color code marking not as specified.
X	X	X	115.	Electrically operated utility and testing equipments not as specified.
X	-	X	116.	Shock sensitive laboratory testing equipment and vibration producing equipment not installed as specified.
X	X	X	117.	Electrical wiring not as specified.
X	-	X	118.	Threaded parts not as specified.
X	-	X	119.	Explosive chemicals, compressed gasses, and other dangerous articles not installed as specified.
X	-	X	120.	Electrical circuitry, including all components and connections, not fungus and moisture resistant as specified.
X	X	X	121.	Fire extinguishers missing or not mounted as specified.
X	-	X	122.	Identification marking and identification plates not as specified.
X	-	X	123.	Shipping data plate not as specified.

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1	2	3	4	
X	-	X	124.	Instruction plates not as specified.
X	-	X	125.	Treatment and painting not as specified.
X	X	X	126.	Technical publications and instructional material missing or not as specified.
X	X	X	127.	Repair parts and maintenance tools missing.
X	-	X	128.	Vehicle weight classification missing or not as specified.
X	X	X	129.	Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Tests shall be performed without shelter and at the climatic conditions existing at the place of test. The mobile laboratory shall operate as specified herein without maintenance other than the supplier's recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the supplier prior to test.

4.5.2.2 Road. The mobile laboratory with all equipment, furnishings and supplies installed, shall be coupled with a government-loaned M-52 tractor truck, FSN 2320-835-8326, and towed over a hard surface road for a minimum of 20 miles at a speed of 50 miles per hour. A circular test track shall not be used for this test. Nonconformance to 3.6 shall constitute failure of this test.

4.5.2.3 Across curbing. The fully loaded and equipped mobile laboratory shall be towed by the M-52 tractor over a section of concrete, asphalt, or solid wood curbing at not less than 5 miles per hour. The dimensions of the curbing shall be 6 inches, plus or minus 1/2 inch, high by 6 inches, plus or minus 1/2 inch, wide. The top corners of the curbing coming into contact with the wheels of the M-52 tractor and the mobile laboratory shall be rounded to a 1-inch, plus or minus 1/8 inch, radius. The curbing shall be anchored to a hard surfaced road and shall be stationary during the test. The full height of the curbing shall protrude above the surface of the road. All wheels of the M-52 tractor and mobile laboratory shall pass over the curbing at a 45-degree, plus or minus 5 degrees, angle to the centerline of the curbing with the road side of the M-52 tractor and mobile laboratory. This test shall be repeated passing all wheels of the M-52 tractor and mobile laboratory over the curbing at a 45-degree, plus or minus 5 degrees, angle to the centerline of the curbing with the curb side of the M-52 tractor and mobile laboratory. Nonconformance to 3.6 shall constitute failure of this test.

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4.5.2.4 Simulated aircraft loading. The simulated aircraft loading test shall be performed on the mobile laboratory with all equipment, furnishings and supplies installed. The trailing end of the mobile laboratory shall be elevated 22 degrees, plus or minus 1 degree, from the horizontal for a period of 5 minutes. The rate of elevation shall be not less than 3 inches per second. The trailing end of the mobile laboratory shall then be returned to ground. Nonconformance to 3.6 shall constitute failure of this test.

4.5.2.5 Pressure testing of utility systems. The mobile laboratory systems shall be pressure tested as indicated:

System	Pressure	Medium	Duration *	Examination
Water	45 psig	Water	30	Visual
Drainage	Gravity	Water	10	Visual
Air	60 psig	Air	20	Soap Solution
Vacuum	0.5 mm Hg	Negative Pressure	15	Manometer
Butane/ Propane	Normal System Pressure	Butane/ Propane	20	Odor and Soap Solution
Oxygen	Normal System Pressure	Oxygen	20	Soap Solution

* Not less than time indicated in minutes.

The above testing and examination shall be conducted for a total of two times. After the first test, the pressures shall be allowed to return to ambient before pressurizing for the second test. Any evidence of leakage shall constitute failure of this test.

4.5.2.6 Functional testing of electrically operated systems and equipment. The electrically operated equipment and electrical systems shall be electrically energized and operated in accordance with the laboratory technical manual.

(a) Time-delay purging system.
(b) Lighting blackout system.
(c) Utility equipment.
(d) Testing equipment.
(e) 24-volt electrical system.
(f) 120/208-volt electrical system.

Nonconformance to 3.10 shall constitute failure of this test.

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4.6 Inspection of preparation for delivery. The packing and marking of each mobile laboratory shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

130. The mobile laboratory not preserved and packaged as specified for level A.
131. Equipment, supplies, accessories, overpack boxes not positioned and secured as specified for level A or C.
132. Pulley grooves not coated with primer as specified for level A.
133. Ferrous metal surfaces requiring a preservative not coated as specified for level A.
134. Tape not as specified for level A.
135. Fluorescent lamps not secured as specified for level A or C.
136. Refrigerant charge on air conditioner not pumped down as specified for level A or C.
137. Air conditioner valves not closed and not tagged as specified for level A or C.
138. Doors, blackout covers, and weather flaps not closed and secured as specified for level A or C.
139. Marking incomplete, incorrect, or missing for level A or C.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified.

5.1.1 Level A.

5.1.1.1 Mobile laboratory. Each mobile laboratory shall be preserved and packaged in accordance with level A (mobile) requirements of MIL-STD-281 and as specified herein (see 5.2.4).

5.1.1.2 Interior. All equipment, supplies, accessories and overpack boxes shall be positioned and secured in designated location as specified in section 3. Overpack boxes shall be closed and secured using 8-penny, steel common nails. Pulley grooves shall be coated with primer conforming to TT-P-664. Exposed, uncoated ferrous metal surfaces requiring the application of a preservative shall be coated with preservatives in accordance with MIL-P-116. Fluorescent lamps shall be secured in position by taping over the slotted section of the holders with minimum 1-inch width tape conforming to PPP-T-60, type IV.

5.1.1.2.1 Air conditioning unit. The refrigerant charge on open type systems shall be pumped down and the appropriate valves closed and capped.

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5.3 Marking. In addition to any caution and deprocessing marking, interior packages and units shall be marked in accordance with MIL-STD-129 and as specified herein. Special marking requirements applicable to unpacked vehicles shall apply. Tool, spare parts and overpack boxes shall contain a list of items contained therein.

5.3.1 Humping precaution. Large signs containing the words "DO NOT HUMP" shall be secured in conspicuous places on each side of the railroad car.

6. NOTES

6.1 Intended use. The mobile laboratory and sustaining equipment covered by this specification are to be used by Armed Forces for testing petroleum and petroleum products under tactical military operations.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of first-produced model (see 3.2).
(c) When a semitrailer is required (see 3.4).
(d) When items of equipment shall be furnished by the supplier (see 3.5).
(e) When screw threads shall be other than as specified (see 3.12).
(f) When the Government will not furnish the fire extinguishers (see 3.15).
(g) When instructional material is not to be furnished by the supplier (see 3.20).
(h) Applicable list of repair parts and maintenance tools (see 3.21).
(i) When Government-furnished property is required (see 3.23 and 6.6).
(j) Weight classification numbers and instructions required (see 3.24 and 6.8).
(k) When the welding process is not to be at the option of the supplier (see 3.25.2).
(l) Level of preservation and packaging required (see 5.1).
(m) The number of mobile laboratories to be loaded on each railcar, the type of railcar required, and the transportation data required (see 5.2.3).

6.3 First-produced mobile laboratory. Any changes or deviations of production mobile laboratories from the approved first-produced mobile laboratory during production will be subject to the approval of the contracting officer. Approval of the first-produced mobile laboratory will not relieve the supplier of his obligation to furnish mobile laboratories conforming to this specification.

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5.3 Marking. In addition to any caution and deprocessing marking, interior packages and units shall be marked in accordance with MIL-STD-129 and as specified herein. Special marking requirements applicable to unpacked vehicles shall apply. Tool, spare parts and overpack boxes shall contain a list of items contained therein.

5.3.1 Humping precaution. Large signs containing the words "DO NOT HUMP" shall be secured in conspicuous places on each side of the railroad car.

6. NOTES

6.1 Intended use. The mobile laboratory and sustaining equipment covered by this specification are to be used by Armed Forces for testing petroleum and petroleum products under tactical military operations.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of first-produced model (see 3.2).
(c) When a semitrailer is required (see 3.4).
(d) When items of equipment shall be furnished by the supplier (see 3.5).
(e) When screw threads shall be other than as specified (see 3.12).
(f) When the Government will not furnish the fire extinguishers (see 3.15).
(g) When instructional material is not to be furnished by the supplier (see 3.20).
(h) Applicable list of repair parts and maintenance tools (see 3.21).
(i) When Government-furnished property is required (see 3.23 and 6.6).
(j) Weight classification numbers and instructions required (see 3.24 and 6.8).
(k) When the welding process is not to be at the option of the supplier (see 3.25.2).
(l) Level of preservation and packaging required (see 5.1).
(m) The number of mobile laboratories to be loaded on each railcar, the type of railcar required, and the transportation data required (see 5.2.3).

6.3 First-produced mobile laboratory. Any changes or deviations of production mobile laboratories from the approved first-produced mobile laboratory during production will be subject to the approval of the contracting officer. Approval of the first-produced mobile laboratory will not relieve the supplier of his obligation to furnish mobile laboratories conforming to this specification.

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6.4 Hazardous materials. Hazardous materials are listed in part 172 of the DOT regulation 49CFR (see 3.13).

6.5 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.15, when applicable.

6.6 Government-furnished property. When specified, the contracting officer should arrange to furnish the property specified in 3.23.

6.7 Government-loaned property. The contracting officer should arrange to loan the property specified in 4.5.2.2.

6.8 Vehicle weight classification. The contracting officer should arrange to furnish vehicle classification forms to the supplier and request the return of the completed forms to the Government. The information thus obtained should be utilized to determine the weight classification number(s) for the laboratory. The supplier should then be requested by the contracting officer to apply the weight classification number(s) to the laboratory in accordance with AR 746-5. The use of a Government-furnished sign kit conforming to MIL-S-40626 should be utilized where applicable (see 3.24).

6.9 Definition. Where an item is specified by trade name and is followed by the words "or equal" or "or equivalent", it is intended to indicate a supplier satisfactory to the Government and a quality level that must be met or surpassed by any supplier of an "or equal" or "or equivalent" item. In order to be considered equal or equivalent, a piece or item must be dimensionally and functionally interchangeable with the piece or item it is to replace.

6.10 List of items. The contracting officer should provide the list of the items specified in SC-6640-97-CL-E02 (see 3.4).

Custodian:

Army - ME

Review activities:

Army - MD, MU

Preparing activity:

Army - ME

Project No. 6640-A622

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION
MIL-L-51050B(ME) Laboratory, Petroleum, Semitrailer Mounted

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO (If "yes" in what way?)

4 REMARKS (Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity - Optional)

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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